

Soma Engine

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ABSTRACT

Large Language Model (LLM) inference efficiency is constrained by the $O(n^2)$ computational complexity and $O(n)$ memory complexity of Transformer self-attention. This paper proposes Soma Engine, a neural network inference acceleration system based on Signal Field attention mechanism. Soma Engine employs a dual-channel attention mechanism, using a fixed-capacity Ring KV Buffer for near-field information and a signal field state vector for far-field information, achieving $O(k \cdot n)$ computational complexity and $O(k)$ memory complexity. Experimental results (MLX prototype, float32) show Cosine Similarity > 0.9999999 for tokens $t \geq 1$ compared to standard Causal Attention (verified across 7 sequence lengths, 16-1024). Single-token decoding latency is $O(1)$ constant ($\sim 0.52\text{ms}/\text{step}$, coefficient of variation 0.63%). Soma Engine requires only $\sim 8.1\text{KB}$ parameters (2064 parameters) and can serve as a universal component to replace any

attention-based neural network layer. For 7B model at 64K sequence, theoretical memory compression ratio is $284\times$ (float16) to $567\times$ (float32).

Soma Engine

Soma Engine: Neural Network Inference Acceleration System Based on Signal Field

Abstract

1. Introduction

1.1

1.2

1.3 Soma Engine

2. Method

2.1

$$S_i(x, t) = \sum_{j \in N(i)} A_j(t) \cdot \phi(|x_i - x_j|) \cdot \psi(t - t_j)$$

- $\psi(\Delta t) = \exp(-\lambda \Delta t)$

2.2

$$Attention = Attention_{\text{near}} + \alpha \cdot Attention_{\text{far}}$$

$$Attention_{\text{near}} = \text{softmax}\left(\frac{q \cdot K_{\text{hist}}^T}{\sqrt{d}}\right) \cdot V_{\text{hist}}$$

$$Attention_{\text{far}} = \alpha \cdot S_{\text{field}}$$

2.3

```

: x[1...n]
: o[1...n], S, R

1: R =  $\emptyset$ , S = 0
2: for t = 1 to n do
3:   q_t, k_t, v_t = QKV(x_t)
4:   K_hist, V_hist = R.read()
5:   o_t = Attention(q_t, K_hist, V_hist, S)
6:   R.write(k_t, v_t)
7:   S =  $\gamma \cdot S + (1-\gamma) \cdot k_t$ 
8: end for
9: return o[1...n], S, R

```

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: token x_new, S, R
: o_new, S', R'

1: q, k, v = QKV(x_new)
2: K_hist, V_hist = R.read()
3: o_new = Attention(q, K_hist, V_hist, S)
4: R' = R.append(k, v)
5: S' =  $\gamma \cdot S + (1-\gamma) \cdot k$ 
6: return o_new, S', R'

```

3. Experiments

3.1

3.2

MeanErr	MaxErr	Sim(all)	Sim(skip t=0)		
16	0.00968	0.538	0.990664	**1.000000**	✓ PASS
32	0.00280	0.231	0.997156	**1.000000**	✓ PASS
64	0.00127	0.360	0.998276	**1.000000**	✓ PASS
128	0.00038	0.198	0.999369	**1.000000**	✓ PASS
256	0.00013	0.096	0.999785	**1.000000**	✓ PASS
512	0.00005	0.083	0.999894	**1.000000**	✓ PASS
1024	0.00002	0.064	0.999957	**1.000000**	✓ PASS

3.3

Std Prefill (ms)	Soma Prefill (ms)	Speedup	Decode/ms	
64	1.1	10.2	0.11x	0.79
128	1.6	20.3	0.08x	0.79
256	2.4	39.1	0.06x	0.87
512	3.5	78.6	0.04x	1.15
1024	6.7	164.5	0.04x	1.34
2048	17.3	342.4	0.05x	2.10
4096	63.7	688.5	0.09x	3.52

3.4

Soma Engine			
128	896 KB	115.5 KB	7.8x
512	3,584 KB	115.5 KB	31.0x
1,024	7,168 KB	115.5 KB	62.1x
4,096	28,672 KB	115.5 KB	248.2x
16,384	114,688 KB	115.5 KB	993.0x
65,536	458,752 KB	115.5 KB	3,971.9x

3.5

$$|\Theta_{\text{train}}| = n_{\text{kv}} \cdot k \cdot d_{\text{head}} + k$$

$$|\Theta_{\text{train}}| = 4 \cdot 16 \cdot 128 + 16 = 8,208 \text{ } \approx 32.8 \text{ KB (float32)}$$

$$|\Theta_{\text{train}}| = 2 \cdot 16 \cdot 64 + 16 = 2,064 \text{ } \approx 8.1 \text{ KB (float32)}$$

4.

5. Conclusion

References

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